संगणक विज्ञान एवं अभियांत्रिकी विभाग
Department of Computer Science & Engineering
भारतीय प्रौद्योगिकी संस्थान जोधपुर
Indian Institute of Technology Jodhpur
एन. एच. 62, नागौर रोड़, करवर, जोधपुर – 342030
N.H. 62, Nagaur Road, Karwar, Jodhpur-342030
दूरभाष संख्या: +91 291 2801252 ईमेल: office_cse@iitj.ac.in

Indian Institute of Technology Jodhpur

M.Tech/ M.Des./ M.Sc./Master's Program Course Curriculum Department of Computer Science and Engineering

Course Structure: Semester-wise distribution of credits for M.Tech. AI students for the batches from Academic Year 2025-26.

Cat.	*Course Number, Course Title*		*L-T-P*	*Credits*		*Cat.*	*Course Number, Course Title*		*L-T-P*	*Credits*
I Semester						**II Semester**				
C	CSL7130	Mathematical Foundations for Computer Science	3-0-0	3		C	CSL7590	Deep Learning	3-0-0	3
C	CSL7610	Artificial Intelligence	3-0-0	3		E	CSL7XXX	Elective	3-0-0	3
C	CSL7620	Machine Learning	3-0-0	3		E	CSL7XXX	Elective	3-0-0	3
E	CSL7XXX	Elective	3-0-0	3		E	CSL7XXX	Elective	3-0-0	3
NG1	HSN7XXX	Non-Graded I	1-0-0	S/X		NG2	HSN7XXX	Non-Graded II	1-0-0	S/X
			Total	**12**					Total	**12**
T	CSS7000	Summer Thesis								8
III Semester						**IV Semester**				
T	CST7010	Thesis		16		T	CST7XXX	Thesis		16
			Total	**16**					Total	**16**

S.No.	*Category*	*Course Category Title*	*Total Courses*	*Total Credits*
1	C	*COMPULSORY*	4	12
2	E	*ELECTIVES*	4	12
3	NG	*Non-Graded*	2	0
3	T	Thesis	**32+8**	32+8
		Total		**64**

Department of CSE

MTech AI Courses

1. The following four courses are compulsory for M.Tech. students of the department:

    1. **CSL7130 Mathematical Foundations for Computer Science**
    2. **CSL7610 Artificial Intelligence**
    3. **CSL7620 Machine Learning**
    4. **CSL7590 Deep Learning**

2. M.Tech. Students shall have the option to choose four courses from the following list of courses:

    1. **CSL7650 Autonomous Systems**
    2. **CSL7770 Speech Understanding**
    3. **CSL7360 Computer Vision**
    4. **CSL7860 Foundation Models and Generative AI**
    5. **CSL7870 Learning on Graphs and its Applications**
    6. **CSL7110 Machine Learning with Big Data**
    7. **CSL7210 ML-DL-Ops**
    8. **CSL7640 Natural Language Understanding**
    9. **CSL7XXX Selected Topics in AI**
    10. **CSL7450 Computer Graphics**
    11. **CSL7XXX Systems for ML**
    12. **CSL7560 Advanced Data Structures and Algorithms**

The syllabus and checklists for the new proposed courses are given below:

Title	**Artificial Intelligence (700)**	Number	CSL7610
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B. Tech., M.Tech., Ph.D.	Type	Compulsory (AI)
Prerequisite	Data Structures and Algorithms	Antirequisite	Artificial Intelligence (300) - CSL 3XX

Objectives

The Instructor will:

1. Cover various paradigms that come under the broad umbrella of AI.

Learning Outcomes

The students are expected to have the ability to:

1. Develop an understanding of where and how AI can be used.

Contents

Introduction: Uninformed search strategies**,** Greedy best-first search, And-Or search, Uniform cost search, A* search, Memory-bounded heuristic search, Local and evolutionary searches (9 Lectures)
Constraint Satisfaction Problems: Backtracking search for CSPs, Local search for CSPs (3 Lectures)
Adversarial Search: Optimal Decision in Games, The minimax algorithm, Alpha-Beta pruning, Expectimax search (4 Lectures)
Knowledge and Reasoning: Propositional Logic, Reasoning Patterns in propositional logic; First order logic: syntax, semantics, Inference in First order logic, unification and lifting, backward chaining, resolution (9 Lectures)
Planning: Situation Calculus, Deductive planning, STRIPES, sub-goal, Partial order planner (3 Lectures)
Bayesian Network, Causality, and Uncertain Reasoning: Probabilistic models, directed and undirected models, inferencing, causality, *Introduction to Probabilistic reasoning* (6 lectures)
Reinforcement Learning: MDP, Policy, Q-value, Passive RL, Active RL, Policy Search (8 Lectures)

Textbook

1. Russel,S., and Norvig,P., (2015), *Artificial Intelligence: A Modern Approach*, 3rd Edition, Prentice Hall

Reference Books

1. Research literature

Self Learning Material

1. Department of Computer Science, University of California, Berkeley, http://www.youtube.com/playlist?list=PLD52D2B739E4D1C5F
2. NPTEL: Artificial Intelligence, https://nptel.ac.in/courses/106105077/

Title	**Machine Learning (700)**	Number	CSL7620
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M.Tech. (CSE, AI, DCS), PhD	Type	Compulsory
Prerequisite	Introduction to Computer Sc., Probability, Statistics and Stochastic Processes, Linear Algebra	Antirequisite	IML, PRML

Objectives

1. To understand various key paradigms for machine learning approaches
2. To familiarize with the mathematical and statistical techniques used in machine learning.
3. To understand and differentiate among various machine learning techniques.

Learning Outcomes

The students are expected to have the ability to:

1. To formulate a machine learning problem
2. Select an appropriate pattern analysis tool for analyzing data in a given feature space.
3. Apply pattern recognition and machine learning techniques such as classification and feature selection to practical applications and detect patterns in the data.

Contents

Fractal I: Supervised Learning

Introduction: Definitions, Datasets for Machine Learning, Different Paradigms of Machine Learning, Data Normalization, Hypothesis Evaluation, VC-Dimensions and Distribution, Bias-Variance Tradeoff, Regression (Linear) (7 Lectures)

Bayes Decision Theory: Bayes decision rule, Minimum error rate classification, Normal density and discriminant functions (5 Lectures)

Parameter Estimation: Maximum Likelihood and Bayesian Parameter Estimation (2 Lectures)

Fractal II: Unsupervised Learning

Discriminative Methods: Distance-based methods, Linear Discriminant Functions, Decision Tree, Random Decision Forest and Boosting (6 Lectures)

Feature Selection and Dimensionality Reduction: PCA, LDA, ICA, SFFS, SBFS (4 Lectures)

Clustering: k-means clustering, Gaussian Mixture Modeling, EM-algorithm (4 Lectures)

Fractal III: Kernels and Neural Networks

Kernel Machines: Kernel Tricks, SVMs (primal and dual forms), K-SVR, K-PCA (6 Lectures)

Artificial Neural Networks: MLP, Backprop, and RBF-Net (4 Lectures)

Foundations of Deep Learning: DNN, CNN, Autoencoders (4 lectures)

Text Book

1. Shalev-Shwartz,S., Ben-David,S., (2014), Understanding Machine Learning: From Theory to Algorithms, Cambridge University Press
2. R. O. Duda, P. E. Hart, D. G. Stork (2000), Pattern Classification, Wiley-Blackwell, 2nd Edition.

Reference Book

1. Mitchell Tom (1997). Machine Learning, Tata McGraw-Hill
2. C. M. BISHOP (2006), Pattern Recognition and Machine Learning, Springer-Verlag New York, 1st Edition.

Self-Learning Material

1. Department of Computer Science, Stanford University, https://see.stanford.edu/Course/CS229

Title	**Deep Learning (700)**	Number	CSL7590
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	M.Tech. 1st Year, Ph.D. 1st Year	Type	Compulsory (AI)
Prerequisite	Machine Learning	Antirequisite	Deep Learning (400) - CSL4xx

Objectives

1. Provide technical details about various recent algorithms and software platforms related to Machine Learning with specific focus on Deep Learning.

Learning Outcomes

Students are expected to have the ability to:

1. Design and program efficient algorithms related to recent machine learning techniques, train models, conduct experiments, and develop real-world DL-based applications and products

Contents

Fractal 1: Foundations of Deep Learning

Deep Networks: CNN, RNN, LSTM, Attention layers, Applications (8 lectures)

Techniques to improve deep networks: DNN Optimization, Regularization, AutoML (6 lectures)

Fractal 2: Representation Learning

Representation Learning: Unsupervised pre-training, transfer learning, and domain adaptation, distributed representation, discovering underlying causes (8 lectures)

Auto-DL: Neural architecture search, network compression, graph neural networks (6 lectures)

Fractal 3: Generative Models

Probabilistic Generative Models: DBN, RBM (3 lectures)

Deep Generative Models: Encoder-Decoder, Variational Autoencoder, Generative Adversarial Network (GAN), Deep Convolutional GAN, Variants and Applications of GANs (11 lectures)

Text Book

1. Goodfellow,I., Bengio.,Y., and Courville,A., (2016), Deep Learning, The MIT Press .

Reference Book

1. Charniak, E. (2019), Introduction to deep learning, The MIT Press.
2. Research literature.

Self Learning Material

1. https://www.deeplearningbook.org/

Title	**Autonomous Systems**	Number	CSL7650
Department/IDRP	CSE	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech, MTech, PhD (All Branches)	Type	Elective
Prerequisite	Basics of Linear Algebra, Probability and Statistics		

Objectives

1. Provide an understanding about autonomous/ semi-autonomous systems like autonomous mobile robots and cars.

Learning Outcomes

1. Understand and use the methodologies to design, model and implementation of autonomous systems for real time applications.

Contents:

Fractal I: Introduction to Autonomous Systems [14 Lectures]

- Introduction to Autonomous Systems, properties of autonomous systems, examples of autonomous systems, degree of autonomy **[3]**
- Abstract Agent Architectures, Environment and agent, reactive, utility-based and behavior based architectures. Concrete agent architectures, logical agents, introduction to Bayesian reasoning **[7]**
- Introduction to real-time systems, real-time environment, distributed solutions, notion of time, fault-tolerance, modeling real-time systems **[4]**

Fractal II: Design of Autonomous vehicles [14 Lectures]

- Architecture of autonomous vehicles, processing pipeline, Vehicle Networks **[1]**
- Sensors for Navigation, GPS, IMU, Lidar, Camera, Visual Odometry, Place Recognition, Extraction based on Range Data **[5]**
- Localization, Noise and Aliasing, Belief Representation, Map Representations, Probabilistic Map based localization, Autonomous Map Building, SLAM **[4]**
- Planning and Reacting, Path Planning and Obstacle Avoidance, Navigation Architectures, Data Fusion **[4]**

Fractal III: Multi-agent systems [14 Lectures]

- Multi-agent systems and organizations, Hierarchical systems & peer-to-peer organization, Openness & scalability, Emergent organization **[2]**
- Communication and coordination in distributed systems, Distributed Planning and Execution, Task sharing, Coordination without communication [3]
- Game Theory, 2-player games, Nash Equilibrium, Strategy, Adversarial search, Iterative game and learning, Pareto-efficiency, Negotiation, Social choice, Mechanism design, Auction **[7]**
- Autonomous systems and human society, ethics and trust in multi-agent systems, explainability **[2]**

Text Books

1. Gerhard Weiss ed. (2013), Multiagent System, Second Edition, MIT Press.
2. Roland Siegwart, Illah Reza Nourbakhsh, Davide Scaramuzza (2018), Introduction to autonomous mobile robots, MIT press.

Reference Books

1. Michael Wooldridge (2009), An introduction to multi-agent systems (2nd ed). Wiley
2. Stuart J. Russell and Peter Norvig (2019), Artificial Intelligence: A Modern Approach, 4th edition,Pearson Press
3. Hermann Kopetz. (2011). Real-time Systems (2nd ed). Kluwer Academic Publishers
4. Choset, H. M., Hutchinson, S., Lynch, K. M., Kantor, G., Burgard, W., Kavraki, L. E., &Thrun, S. (2005). Principles of robot motion: theory, algorithms, and implementation. MIT press.

Other study materials

Research papers: (to be shared in class)

Online Course Material:

https://www.edx.org/course/autonomous-mobile-robots

Course Title	**Speech Understanding**	Course No.	**CSL7770**
Department	CSE / EE	Structure (L-T-P-C)	3-0-0 [3]
Offered for	B.Tech CSE, AI&DS	Type	Elective
Prerequisite	PRML/IML/ML	Antirequisite	None

Objectives

1. To provide insights into fundamental concepts and algorithms related to speech processing and understanding
2. Impart working expertise by introducing practical problems.

Learning Outcomes

Students will have the ability to:

1. Build a speech recognition system
2. Design and implement basic speech based application

Contents

Introduction to Speech processing: Digitization and Recording of speech signal, Review of Digital Signal Processing Concepts, Human Speech production, Acoustic Phonetics and Articulatory Phonetics, Different categories speech sounds and Location of sounds in the acoustic waveform and spectrograms. (14 Lectures)

Speech recognition*:* Analysis and Synthesis of Pole-Zero Speech Models, Short-Time Fourier Transform, Analysis:- FT view and Filtering view, Synthesis:-Filter bank summation (FBS) Method and OLA Method, Features Extraction, Extraction of Fundamental frequency, Speech Enhancement, Clustering and Gaussian Mixture models, Speaker Recognition.(14 Lectures)

Speech based applications: HMM and Neural models for speech recognition, Speech generation, Question answering, Dialogue systems, Other Speech based Applications.(14 Lectures)

Text Books

1. T.F. QUATIERI (2002), Discrete-Time Speech Signal Processing, Prentice-Hall, New Jersey.
2. D. JURAFSKY, J.H. MARTIN, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition (3rd Edition Draft), 2019.

Reference Books

1. Y. GOLDBERG (2016), A Primer on Neural Network Models for Natural Language Processing, Journal of Artificial Intelligence Research.
2. I. GOODFELLOW, Y. BENGIO, A. COURVILLE (2016), Deep Learning, The MIT Press, 1st Edition.
3. S.K. PATRA (2011), Digital Signal Processing: A Computer-Based Approach, McGraw-Hill, 4th Edition.

Title	**Computer Vision**	Course No.	CSL7360
Department	Computer Science and Engineering, AI&DS	Structure (L-T-P [C])	3–0–0 [3]
Offered for	B.Tech., M.Tech., Ph.D.	Type	Elective
Prerequisite	Linear Algebra		

Objectives

The Instructor will:

1. Provide insights into fundamental concepts and algorithms behind some of the remarkable success of Computer Vision
2. Impart working expertise by means of programming assignments and a project

Learning Outcomes

The students are expected to have the ability to:

1. Learn and appreciate the usage and implications of various Computer Vision techniques in real-world scenarios
2. Design and implement basic applications of Computer Vision

Contents

Introduction*:* The Three R's - Recognition, Reconstruction, Reorganization (1 Lecture)
Fundamentals: Formation, Filtering, Transformation, Alignment, Color (5 Lectures)
Image Restoration*:* Spatial Processing and Wavelet-based Processing (5 Lectures)
Geometry: Homography, Warping, Epipolar Geometry, Stereo, Structure from Motion, Optical flow (9 Lectures)
Segmentation: Key point Extraction, Region Segmentation (e.g., boosting, graph-cut and level-set), RANSAC (6 Lectures)
Feature Description and Matching: Key-point Description, handcrafted feature extraction (SIFT, LBP) (3 Lectures)
Deep Learning based Segmentation and Recognition: DL-based Object detection (e.g. Mask-RCNN, YOLO), Semantic Segmentation, Convolutional Neural Network (CNN) based approaches to visual recognition (9 Lectures)
Applications: Multimodal and Multitask Applications (4 Lectures)

Textbooks

1. R. HARTLEY, A. ZISSERMAN (2004), Multiple View Geometry in Computer Vision, Cambridge University Press, 2nd Edition.
2. R. SZELISKI, (2010), Computer Vision: Algorithms and Applications, Springer-Verlag London.

Reference Books

1. Research literature

Title	**Foundation Models and Generative AI**	Number	CSL7860
Department	Computer Science and Engineering	L-T-P [C]	3-0-0 [3]
Offered for	B. Tech, MTech, PhD (All Branches)	Type	Elective
Prerequisite	IML/PRML/ML and Deep Learning	Antirequisite	None

Objectives

1. This course aims to equip students with a deep understanding of foundation models, focusing on its fundamental concepts and core models. Students will learn about the concept and impact of foundation models within the generative AI domain. The course will explore the extensive capabilities of pre-trained models and their applications in AI-powered solutions. Additionally, students will gain practical insights into the features, capabilities, and real-world applications of various generative AI platforms. Students will be able to critically assess and effectively implement generative AI technologies in diverse contexts.

Learning Outcomes

1. The students will be able to understand the principles and formulations of foundation models,
2. The students will be able to conceptualize and design new foundation models, and utilize them for different applications and tasks,
3. The students will be able to analyze foundation models from different perspectives such as performance, computational requirements, explainability and interpretability,
4. The students will be able to perform inference and fine-tune foundation models.

Contents

Introduction to Foundation Models:

- Transformers & Attention : Introduction to Transformers, Self-Attention, Cross-Attention, Perceiver: Arbitrary IO with transformers, Self Attention & Non-Parametric Transformers (6 lectures)
- Diffusion : Sampling, Diffusion Models, Latent Diffusion Models (LDM) (6 lectures)
- Applications : LLMs, LVMs and Applications in Vision (2 lectures)

Foundation Model Architectures:

- Foundation Model Architectures: Transformers in Language including GPT, BERT, RetNet, State Space Models (4 lectures)
- Transformers in Vision: ViT, MLP-Mixer, Conformer, Vision-Language(VL) models, CLIP (4 lectures)
- Architectures: Dual-Encoder, Fusion, Encoder-Decoder, Adapted LLM, Unified Architectures (6 lectures)

Multi-Modal & Generative Foundation Models:

- Training & Inference: Training Objectives, Contrastive Learning, Efficient Inference Techniques, Pre-training, Fine-tuning & Parameter Efficient Fine-tuning (LoRA, QLoRa), Flash Attention, Retrieval Augmented Generation (RAG) (8 lectures)

Evaluation & Benchmarking:

- Evaluation Protocols and Standard Benchmarks, Hallucinations, Bias & Fairness, Privacy, Memorization, Machine Unlearning, Prompt Engineering (6 lectures)

Evaluation Plan:

- 3 exams and quizzes = 15+15+15+5 = 5o%
- Coding Project with three continuous evaluation components: 1o+1o+2o = 4o%
- Paper Reviews and class presentation: 5+5 = 1o%

Textbook & Reference Material

1. Foundation Models for Natural Language Processing - Pre-trained Language Models Integrating Media, Gerhard Paaß , Sven Giesselbach, Springer 2o23.
2. Research papers will be shared in class

Course Title	**Learning on Graphs and its Applications**	Course No.	**CSL7870**
Department	Computer Science and Engineering	Structure (L-T-P [C])	3-0-0 [3]
Offered for	B.Tech, MTech, PhD	Type	Elective
Pre-requisites	IML/PRML for UG	Preferred Knowledge	Basic ML

Objectives
Complex data can be modeled as a graph of relationships and interactions between objects. Graph data structures can be processed by algorithms like neural networks to perform tasks such as classification, clustering, and regression.This course investigates the computational, algorithmic, and modeling challenges specific to graph analysis. By studying underlying graph structures, students will gain mastery of Graph Neural Network techniques that can enhance prediction and uncover insights across diverse networks.

Learning Outcomes
The students are expected to have the ability to:
1. Understand the applications related to graph and graph representation learning
2. Formulate real-world problems with relational data and use GNN for prediction, classification and other tasks on graph data structures

Contents
Introduction: Complex Systems, Graph Data Structure, Graph Representations, Spectral graph theory , Real world networks and applications. (6 Lectures)

Graph Representation Learning: Node Embeddings , Graph Convolutional Networks, Design Space, Inductive Representation Learning, (6 Lectures)

Graph Attention Networks, Hierarchical Graph Representation Learning, Graph Embeddings, Over smoothing Issues, Expressiveness of GNN, GNN vs CNN (8 Lectures)

GNN Applications: Heterogeneous and Multilayer graphs, Knowledge graphs, Reasoning over knowledge graphs, Graphs and LLMs (8 Lectures)

Neural subgraph matching, GCN for Recommender Systems, Node Classification, Graph Classification, (6 Lectures)

Community Detection, Link Prediction and Causality. (4 Lectures)

Large Scale Graphs: Dealing with large graphs, Cluster-GCN, Simplifying Graph Convolutional Networks (4 Lectures)

Textbook
1. Graph Representation Learning by William L. Hamilton
2. Networks, Crowds, and Markets: Reasoning About a Highly Connected World , by David Easley and Jon Kleinberg, (Cambridge University Press - Sep 2010) -- Pre-publication draft available online.

Reference Books
1. Network Science , by Albert-Laszlo Barabasi, (Cambridge University Press - August 2016) -- freely available under the Creative Commons license.

Online Materials:
1. http://web.stanford.edu/class/cs224w/
2. https://youtube.com/playlist?list=PLoROMvodv4rPLKxIpqhjhPgdQy7imNkDn&si=39XL6M_ZzZSSD10j

Title	**Machine Learning with Big Data**	Number	CSL7110
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech, M.Tech., PhD	Type	Elective
Prerequisite	Machine Learning / PRML	Antirequisite	None

Objectives

The Instructor will:

1. Provide an understanding of the role of big data in the real-world scenarios
2. Provide technical details about various algorithms and software/hardware tools/platforms related to big data

Learning Outcomes

Students are expected to have the ability to:

1. Develop an understanding of big data in the modern context, and independently work on problems relating to big-data
2. Design and program efficient algorithms for big data from the perspective of a project

Contents

Introduction: What is big data, Unreasonable effectiveness of data (1 lecture)
Streaming algorithms: Streaming Naive Bayes, Stream and sort (2 lectures)
Platforms for learning from big data: MapReduce, New Software Stack, Large Scale File System Organization (5 lectures)
Nearest Neighbour Search, Jaccardi Similarity of Sets, Similarity of Documents, Locality Sensitive Hashing, The Stream Data Model (4 lectures)
Randomized methods: Clustering, Hashing, Sketching, Scalable stochastic gradient descent (3 lectures)
Frequent Itemsets: The Market Basket Model, A-Priori Algorithm, Handling larger datasets in Main Memory, Limited-Pass Algorithms, Counting Frequent Items in a Stream (6 lectures)
Parameter Servers: Introduction, Abstraction, Parameter Cache Synchronization, Asynchronous execution, Model Parallel Examples (3 lectures)
Graph-based methods (6 lectures)
Page Rank, Topic Sensitive Page Rank, Approaches to Page Rank iteration, Link Spam, Semi-supervised learning, Scalable link analysis, Models for Recommendation Systems, Social Networks as Graphs (9 lectures)
Large-scale Machine Learning with CPUs and GPUs (3 lectures)

Textbook

1. Leskovec,J., Rajaraman,A., Ullman,J., (2014), *Mining of Massive Datasets*, 2nd Edition, Cambridge University Press
2. Bekkerman,R., Bilenko,M., Langford,J., (2011), *Scaling Up Machine Learning*, Cambridge University Press

Reference Books

1. Research literature

Self Learning Material

1. Department of Machine Learning, Carnegie Mellon University, Machine Learning with Large Datasets Course
2. Department of Computer Science, University of California, Berkeley, Scalable Machine Learning
3. ETH Zurich, Data Mining: Machine Learning from Large Datasets

Title	**ML-DL-Ops**	Number	**CSL7210**
Department	Computer Science	L-T-P [C]	2–0–2 [3]
Offered for	B.Tech., M.Tech., Ph.D.	Type	Elective
Prerequisite	Machine Learning or PRML or IML	Antirequisite	None

Objectives

1. To equip students with the ability to design and implement ML/DL systems for real-world applications.
2. To develop expertise in building deployable, reliable, and scalable ML/DL products that meet production standards.

Learning Outcomes

By the end of the course, students will be able to:

1. Design ML/DL systems that solve practical problems and operate in production environments.
2. Diagnose and troubleshoot system faults, identifying root causes effectively.
3. Evaluate and select appropriate frameworks, infrastructures, and trade-offs for scalable deployment.
4. Ensure reproducibility and reliability in ML/DL workflows through experiment tracking and versioning.
5. Analyze trade-offs in system design, balancing accuracy with latency, cost, scalability, and maintainability.
6. Apply techniques for distributed training, mixed precision, and performance optimization.
7. Monitor, update, and maintain deployed ML/DL systems to ensure robustness and fault tolerance.
8. Critically assess emerging trends in ML/DL system design for next-generation applications.

Contents

- Foundations of ML/DL Systems: Data to Deployment Pipeline, Challenges, Lab-to-Production Failures, DevOps to MLOps. (2 lectures)
- Infrastructure & Tools: Virtual Environments, Accelerators, Compilers, Containers, Kubernetes, Reproducibility, Reusability (2 lectures) (2 labs)
- Model Development & Optimization: ML/DL Data Structure and Data Processing, Exploratory Data Analysis and Feature Engineering, Model Development, Hyperparameter Tuning, Model Visualization, Experiment Tracking, Quantization, Mixed Precision, Distributed Training, Large Data Requirement, Inference time optimizations methods like vLLM, TensorRT, ONNX, Model compression methods including pruning and knowledge distillation, libraries and frameworks for model training and inference such as keras, tensorflow, pytorch. (8 lectures) (4 labs)
- Deployment & Operations: Platforms, Versioning, Monitoring, Updating, Troubleshooting, Deployment and Inference Pipelines: CI/CD pipelines, Training Pipeline v/s Inference Pipeline. (4 lectures) (2 labs)
- Performance & Scalability: Trade-offs, Profi ling, Client-Level Inference, Latency Optimization. (4 lectures) (2 labs)
- Advanced Topics: Next-gen DL Systems, Architecture-aware Training, System-level Optimization, LoRA, LLMOps. (4 lectures) (2 labs)
- Ethics & Reliability: Bias, Fairness, Security, Privacy, and Compliance in Production ML Systems, Compliance & Governance: Regulatory requirements (GDPR, HIPAA, AI Act), Explainability. (2 lectures) (1 lab)

The evaluation components should contain a Capstone Project: End-to-End ML/DL System Design with Real-World Deployment with (i) dataset preparation & versioning, (ii) training + experiment tracking, (iii) deployment pipeline, (iv) monitoring + updating. The evaluation components should also include practical assignments in the theory component as well.

Textbooks

1. Online materials and sysML level papers

Self Learning Material

1. https://stanford-cs329s.github.io/
2. https://ucbrise.github.io/cs294-ai-sys-sp19/
3. https://ckaestne.github.io/seai/
4. https://www.coursera.org/specializations/machine-learning-engineering-for-production-mlops#courses
5. MLflow – for MLOps https://mlfl ow.org/docs/latest/ml/
6. Tensor Board – for training visualization https://www.tensorflow.org/tensorboard
7. Jupyterlab notebooks for model training and experimentation. This is also open source https://jupyter.org/.

Title	**Natural Language Understanding**	Number	CSL7640
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech, M.Tech., PhD	Type	Elective
Prerequisite	Deep Learning	Antirequisite	None

Objectives

The Instructor will:

1. To provide insights into fundamental concepts and algorithms related to Natural Language Understanding
2. Impart working expertise by introducing practical problems.

Learning Outcomes

The students are expected to have the ability to:

1. Formulate natural language understanding tasks
2. Design and implement basic applications of NLU

Contents

Traditional NLU: Introduction to NLU, Motivation, Morphology, Parts-of-Speech, Language Models, Word Sense Disambiguation, Anaphora Resolution, Basics of Supervised and Semi-supervised Learning for NLU, Hidden Markov Models for language modeling, EM Algorithm, Structured Prediction, Dependency Parsing, Topic Models, Semantic Parsing, Sentiment analysis. (14 Lectures)

Deep Learning for NLU: Intro to Neural NLU, Word Vector representations, Neural Networks and backpropagation -- for named entity recognition, Practical tips: gradient checks, overfitting, regularization, activation functions, Recurrent neural networks -- for language modeling and other tasks, GRUs and LSTMs -- for machine translation, Recursive neural networks -- for parsing, Convolutional neural networks -- for sentence classification, Question answering and dialogue system, Graph Neural Network for NLU, Natural Language Generation, Analysis and Interpretability of Neural NLU. (22 Lectures) Knowledge Graphs: Knowledge graph embedding techniques, Inference on knowledge graphs. (6 Lectures)

Textbook

1. C. MANNING, H. SCHÜTZE (1999), Foundations of Statistical Natural Language Processing, MIT Press.
2. D. JURAFSKY, J.H. MARTIN, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition (3rd Edition Draft), 2019.

Reference Books:

1. E. BENDER (2013), Linguistic Fundamentals for NLP, Morgan Claypool Publishers..
2. J. ALLEN (1995), Natural Language Understanding, Pearson Education, 1995.
3. Research Literature.

Self-Learning Material

1. http://web.stanford.edu/class/cs224n/index.html#schedule (Deep learning for NLP)

Title	**Selected Topics in Artificial Intelligence - III**	Number	CSL7XX0
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [1]
Offered for	B.Tech., M.Tech., PhD	Type	Elective
Prerequisite	Decided by the instructor		

Objectives
The Instructor will:
1. Expose the students to the latest upcoming fields in the area of artificial intelligence

Learning Outcomes
The students are expected to have the ability to:
1. Apply the knowledge of recent topics to specific research areas in the field of artificial intelligence

Contents
The topic clouds for the course include contemporary topics in artificial intelligence and may be updated according to the instructor.

Textbook
Relevant Textbook and/or research papers to be announced by the instructor.

Self-Learning Material
Relevant Textbook and/or research papers to be announced by the instructor.

Preparatory Course Material
Relevant Textbook and/or research papers to be announced by the instructor.

Title	**Computer Graphics (700)**	Number	CSL7450
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech, M.Tech., Ph.D.	Type	Elective
Prerequisite	None	Antirequisite	Computer Graphics (400) - CSL4xx

Objectives

The Instructor will:

1. Provide a thorough introduction to computer graphics techniques, focusing on 2D and 3D modelling, image synthesis and rendering

Learning Outcomes

The students are expected to have the ability to:

1. Explain and create interactive graphics application
2. Implement graphics primitives
3. Synthesize and render images for animation and visualization

Contents

CSL7xx1 Introduction to Graphical Primitives 1-0-0 [1]

Introduction: Overview of computer graphics, representing pictures, preparing, presenting & interacting with pictures for presentations; Scan conversion: 2D Geometric Primitives; Area Filling algorithms. Clipping algorithms, Anti Aliasing

Transformations and viewing: 2D and 3D transformations, Matrix representations & homogeneous coordinates, Viewing pipeline, Window to viewport coordinate transformation, clipping operations, viewport clipping, 3D viewing.

CSL7xx2 Graphical Object Representation 1-0-0 [1]

Curves and Surfaces: Conics, parametric and non-parametric forms; Bezier (Bernstein Polynomials) Curves, Cubic-Splines, B-Splines; Quadratic surfaces, Bezier surfaces and NURBS, 3-D modelling.

CSL7xx3 Graphics Rendering 1-0-0 [1]

Hidden surfaces: Depth comparison, Z-buffer algorithm, Back face detection, BSP tree method, the Printer's algorithm, scan-line algorithm; Hidden line elimination, wire frame methods, fractal - geometry.

Color & shading models: Phong's shading model, Gouraud shading, Shadows and background, Color models, Photo-realistic rendering, Animation and OpenGL primitives: Functions, pipeline, sample programs for drawing 2-D, 3-D objects; event handling and view manipulation, Introduction to GPU and animation

Textbook

1. J. D. Foley, A. Van Dam, S. K. Feiner and J. F. Hughes, Computer Graphics; Principles and practice, Addison Wesley, 2nd Edition in C, 1997.
2. D. F. Rogers and J. A. Adams, Mathematical elements for Computer Graphics, McGraw-Hill, 2nd Edition, 1990.

Self-Learning Material

1. Blender: *https://www.blender.org/download/*
2. *OpenGL: http://www.opengl-tutorial.org/*

Preparatory Course Material

1.Department of Computer Science and Engineering, Indian Institute of Technology Madras, https://nptel.ac.in/courses/106106090/

Title	**Advanced Data Structures & Algorithms**	Number	**CSL7560**
Department	Computer Science	L-T-P [C]	3–0–0 [3]
Offered for	B.Tech, M.Tech. CSE and M.Tech. AI, Ph.D.	Type	Core
Prerequisite	Algorithm Design and Analysis, Data Structures, Maths for Computing	Antirequisite	None

Objectives

1. The objective of the course is to introduce several advanced algorithmic techniques and data structures.
2. Implementation of various advanced data structure and algorithms to solve real life problems

Learning Outcomes

Students are expected to have the ability to:

1. Learn advanced data structure techniques
2. Learn a new set of techniques to cope with NP-hard problems.
3. Identify novel and significant open research questions in the field.
4. Implementation of various advanced data structure and algorithms to solve real life problems
5. Understand the intractability of problems.

Contents

Data Structures and Algorithms Techniques (Fractals I and II)

Advanced Data Structures [10 lectures]: Red-black trees, Interval trees, Dynamic order statistics, Binomial heap, Fibonacci Heap [5 lectures], Tries, Splay Trees, B-Trees, Hashing - disjoint set, union find [5 lectures].

Network Flows [5 lectures]: Maximum Flow problem, Ford-Fulkerson algorithm, Max flow min cut theorem, augmenting paths, applications to bipartite matching problem and disjoint paths in directed and undirected graphs.

Amortized Analysis [3 lectures]: Aggregate Analysis, Accounting method, Potential method.

Dynamic Programming [6 lectures]: Knapsack, Independent Set of Trees, Interval Scheduling, Set Cover, Permutation Problems, Partition Problems, String matching: naive method, Knuth-Morris-Pratt algorithm.

NP-completeness and Reductions [8 lectures]: Complexity classes P, NP, co-NP, NP-complete, NP-Hard, Example reductions between problems.

Fractal III: Advanced Algorithmic Techniques

Advanced Algorithmic Techniques [10 lectures]: Approximation Algorithms: Greedy Algorithm – Load Balancing, Set Cover [2 Lectures]; The Pricing Method: Vertex Cover, Knapsack [3 Lectures], Randomized Algorithms [5 Lectures].

Textbooks

1. Jon Kleinberg, Eva Tardos (2005), Algorithm Design, Pearson Education, 1st Edition.
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms.

Self Learning Material

1. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-046j-introduction-to-algorithms-sma-5503-fall-2005/video-lectures/
2. https://nptel.ac.in/courses/106/101/106101060/
3. https://nptel.ac.in/courses/106/104/106104019/

Title	**Systems for Machine Learning**	Number	CSL7XX0
Department	Computer Science and Engineering	L-T-P [C]	3–0–0 [3]
Offered for	PhD, MTech, Executive MTech, UG-4	Type	Elective
Prerequisite	Course in Machine Learning/Deep Learning		

Objectives

The Instructor will:

1. Provide full background of the emerging area lying in the intersection of ML/AI, data management, and systems.
2. Provide knowledge on the landscape and evolution of systems and designs that power modern data science applications, large complex datasets, enterprise analytics, recommendation systems, and social media analytics.

Learning Outcomes

The students are expected to have the ability to:

1. Learn core systems topics that span the entire lifecycle of ML based data analytics.
2. Develop skills for data sourcing and preparation, programming models and systems, scalable ML model building, and fast ML deployment.
3. Build systems that incorporate the correct performance landscape, optimization, and trade-offs between frameworks and compute infrastructure.

Contents

Hardware for Machine Learning

Basics of Computer Organization and Operating Systems, Basics of high performance computing in the cloud, GPU Architecture, GPU Programming Models, GPU Applications design, TPU architecture, Domain Specific Architectures. (10 lectures)

ML Training at Scale

Basics of parallel and scalable data processing, parallelism basics, scalable data access, data parallelism, dataflow systems, classical ML training at scale, deep learning systems, feature engineering and model selection systems, techniques in model compression. (20 lectures)

ML Platforms

Frameworks for Parallel and Distributed Training, DL Compilers, Design of ML Pipelines, Scheduling, ML Deployment, Case studies using Horovod, Distributed PyTorch, TensorFlow Extended (TFX), MLFlow, Michelangelo. (12 lectures)

Textbook

1. Boehm M, Kumar A, Yang J, (2020) Data Management in Machine Learning Systems. Morgan Claypool
2. Ramakrishnan R, Gehrke J, (2002) Data Management Systems (3rd Edition). Irwin Computer Science.

References/Self-Learning Material

1. Computer Organization and Design (5th edition), by David Patterson and John Hennessy
2. Programming Massively Parallel Processors, (3rd Edition) by Wen Mei Hwu and David B Kirk.
3. ML Framework guides from respective websites.
4. Research papers.